HRIDAYADARPANA

Comprehensive Cardiovascular Risk Assessment & Clinical Prevention Report

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PATIENT INFORMATION & BIOMETRIC PROFILE

PATIENT NAME **allah khan**	AGE / GENDER **45 yrs / MALE**	PATIENT ID **195319fe-d8b**
BLOOD PRESSURE **125 / 80 mmHg**	RESTING HEART RATE **75 bpm**	BODY MASS INDEX (BMI) **24.5 kg/m2**
TOTAL CHOLESTEROL **190 mg/dL**	FASTING BLOOD SUGAR **95 mg/dL**	SMOKING STATUS **Active Smoker**

EXECUTIVE CLINICAL DIAGNOSTIC INTERPRETATION

Cardiovascular risk score calculated at 30% (MODERATE Risk category). Primary contributing metrics include blood pressure (125/80 mmHg), cholesterol (190 mg/dL), and clinical lifestyle parameters.

CARDIOVASCULAR RISK SCORE & MULTI-HORIZON PROGNOSIS

OVERALL CARDIOVASCULAR RISK ASSESSMENT

MODERATE RISK (30.0%)

Confidence: 92% | Ensemble Engine: v2.1-ClinicalFallback

PROJECTED RISK HORIZONS

30-Day Acute Probability:	**8.4%**
1-Year MACE Probability:	**19.5%**
5-Year Long-Term Probability:	**28.5%**

MULTI-MODEL PREDICTIVE ASSESSMENT

MODEL / PREDICTIVE DOMAIN	MODEL FINDING / METRIC	RISK CLASSIFICATION
Framingham 10-Yr CHD Model	25.5% 10-Yr Likelihood	MODERATE
Kaggle Lifestyle Cardio Ensemble	30.0% Calibrated Risk	MODERATE
Heart Failure Reserve & Strain	Compensated Function	LOW
Stroke Predictive Likelihood	19.9% Cerebrovascular Risk	LOW
PhysioNet ECG Status	NORMAL	NORMAL

EXPLAINABLE AI (SHAP) - KEY RISK DRIVERS

SHAP isolates each biomarker's mathematical contribution driving this patient's cardiovascular risk score up or down.

PHYSIOLOGICAL FACTOR	PATIENT RECORDED VALUE	DIRECTION	IMPACT WEIGHT
Systolic Blood Pressure	125 mmHg	Protective / Lower (-)	+8% Weight
Serum Cholesterol	190 mg/dL	Protective / Lower (-)	+6% Weight
Age & Demographics	45 yrs	Protective / Lower (-)	+18% Weight
Active Smoking	Yes	Increases Risk (+)	+25% Weight
Exercise Frequency	3 days/wk	Protective / Lower (-)	+15% Weight

CONTINUED EVALUATION: Personalized Prevention Protocols, nutrition guidance, and clinical consultation schedules continue on Page 2.

DIGITAL TWIN SIMULATION STATE

SIMULATED STATE: NORMAL (MODERATE SEVERITY)

HEMODYNAMIC CADENCE: Normal synchronized physiological sinus rhythm (1.0x amplitude)

Your latest Risk Scan indicates optimal cardiovascular parameters. The Digital Twin is operating in baseline synchronized mode.

PATIENT-SPECIFIC PRIORITIZED PREVENTION PROTOCOL

Recommendations are prioritized based on this patient's actual Risk Scan metrics and modifiable risk drivers.

HIGH PRIORITY — SMOKING CESSATION — KEY SHAP DRIVER

Urgent Tobacco Cessation Protocol

WHY THIS IS RECOMMENDED (BASED ON RISK SCAN)

Active smoking was identified as one of your highest modifiable risk drivers. Cigarette toxins cause immediate coronary endothelial injury and double the incidence of coronary events.

RECOMMENDED CLINICAL ACTION

Develop a 30-day smoking cessation plan. Inquire with your doctor regarding clinically validated nicotine replacement therapy (patches/gum) or non-nicotine oral medications.

LOW PRIORITY — BLOOD PRESSURE

Maintain Healthy Blood Pressure

WHY THIS IS RECOMMENDED (BASED ON RISK SCAN)

Your blood pressure is within optimal physiological range (125/80 mmHg). Sustaining this baseline preserves vascular elasticity.

RECOMMENDED CLINICAL ACTION

Continue balanced dietary habits with regular hydration and quarterly preventative blood pressure checkups.

LOW PRIORITY — LIPID MANAGEMENT

Sustain Cardioprotective Dietary Habits

WHY THIS IS RECOMMENDED (BASED ON RISK SCAN)

Your serum cholesterol (190 mg/dL) is within desirable range. Maintaining healthy dietary fats supports continuous arterial clarity.

RECOMMENDED CLINICAL ACTION

Preserve your balanced intake of healthy mono- and polyunsaturated lipids with whole grains and fresh produce.

PROGRESSIVE PHYSICAL ACTIVITY PLAN

Cardiorespiratory Endurance & Vascular Fitness - Status: Regularly Active (3 workout days/week)

Initial Target: 30 minutes of moderate aerobic exercise (brisk walking, cycling, swimming) 5 days/week.

Progression: Incorporate 2 days of functional bodyweight or light resistance exercises with dynamic stretching.

Safety Precaution: Monitor exertion; remain within 60-75% of maximum predicted heart rate.

URGENT TOBACCO CESSATION PRIORITY

Active smoking is your highest modifiable coronary risk driver. Establish a 30-day quit date with physician-guided cessation support.

CONTINUED: Detailed Quantified Nutrition Targets, Meal Structure, and Clinical Follow-Up Schedules on Page 3.

PERSONALIZED QUANTIFIED NUTRITION TARGETS

NUTRIENT / METRIC	EVIDENCE-BASED TARGET	CLINICAL GUIDELINES & PRIORITIES
Energy Target	**2,417 - 2,617 kcal/day**	Calibrated to patient BMR, physical activity, and BMI deficit
Dietary Fibre	**28 - 32 g/day (balanced whole grain & produce fibre)**	Prioritize soluble fractions (oats, legumes, pectin, chia seeds)
Carbohydrates	**272 - 327 g/day (50 - 55% of daily energy)**	Intact whole grains, legumes, and low-glycemic vegetables
Dietary Sodium	**< 2,200 - 2,300 mg/day (Standard cardioprotective limit)**	Strict DASH limit; avoid cured meats, bouillon, and processed snacks
Saturated Fat	**< 7 - 8% of daily caloric intake (< 16 - 18 g/day)**	Keep < 11-14g/day; replace with cold-pressed olive oil & nuts
Proteins	**75 - 94 g/day (approx. 1.0 - 1.2 g/kg body weight)**	Fatty fish (salmon, sardines) 2x/wk, skinless poultry, lentils
Added Sugars	**< 25 g/day (< 5% of daily energy; WHO cardioprotective guideline)**	Zero sugar-sweetened beverages or refined bakery syrups
Fruit & Produce	**5 - 7 servings/day (colorful cruciferous, dark greens, and low-**	Cruciferous vegetables, dark greens, and antioxidant-rich berries
Clean Hydration	**2.0 - 2.5 L/day of clean water (adjust for sweat and physical exertion)**	Pure clean water; adjust for climate and physical sweat

WHAT TO EAT MORE OF (CARDIOPROTECTIVE)

+ Abundant dark leafy greens, cruciferous vegetables, and seasonal berries
+ Extra virgin olive oil as the primary culinary fat
+ Raw nuts, seeds, and minimally processed whole grains

WHAT TO STRICTLY LIMIT (ATHEROGENIC)

- Ultra-processed snack foods and deep-fried preparations
- Refined sugars, high-fructose corn syrups, and sweetened bakery items
- Excessive dietary sodium and artificial trans-fat shortenings

SUGGESTED DAILY MEAL STRUCTURE

BREAKFAST

* Slow-digesting complex beta-gluca…
* Topped with 1 tbsp ground flaxseed…
* Lean protein source

LUNCH

* Large mixed green bowl (spinach,...
* 120g grilled skinless chicken br
* Olive oil vinaigrette

MID-DAY SNACK

* Small handful of raw unsalted wa
* Crisp apple slices
* Keep between 120 - 180 kcal; strict…

DINNER

* Steamed broccoli, asparagus spea...
* 120g baked white fish, lentil-ve
* 1/2 cup steamed whole barley or…

RECOMMENDED CLINICAL CONSULTATION TIMELINE
Discuss these risk factor trends with your primary care provider during your next routine visit (recommended within 3 months).

CLINICAL DECISION-SUPPORT NOTICE: This document is generated for cardiovascular risk assessment and preventive decision support. It is not an official medical diagnosis and does not replace in-person diagnostic evaluation by a certified physician or cardiologist. Individualized nutritional and caloric targets should be confirmed with a qualified healthcare professional or registered dietitian. Do not modify or discontinue prescribed medical treatments without consulting your healthcare provider.